BEING A LOCAL COMPLETE INTERSECTION IS A PROPERTY OF THE FORMAL FIBRES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The presentation of a finitely generated field extension as the fraction field of a global complete intersection. Invoked inside the counted proof (source lines 4498-4501). Supplies the auxiliary algebra over which the base change is a relative global complete intersection.
- The permanence of the complete intersection property under a relative global complete intersection. Invoked inside the counted proof (source lines 4502-4504 and 4519-4520). Carries the property through the base change and through the regular extension in part (C).

Source reference. Original reference algebra-lemma-colimit-syntomic: sources/stacks-dualizing/pinned-original-source.tex, source lines 4426–4426. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-colimit-syntomic, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference dpa-definition-lci: sources/stacks-dualizing/pinned-original-source.tex, source lines 4492–4492. Navigation pointer only: the counted text refers to the Stacks item dpa-definition-lci, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference dpa-lemma-avramov: sources/stacks-dualizing/pinned-original-source.tex, source lines 4504–4504. Navigation pointer only: the counted

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Source reference. Original reference more-algebra-section-properties-formal-fibres: sources/stacks-dualizing/pinned-original-source.tex, source lines 4385–4385. Navigation pointer only: the counted text refers to the Stacks item more-algebra-section-properties-formal-fibres, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Properties (A), (B), (C), (D), and (E) of More on Algebra, Section more-algebra-section-properties-formal-fibres hold for $P(k \to R) =$"R is a local complete intersection". See Divided Power Algebra, Definition dpa-definition-lci.*

Proof. Part (A). Let K/k be a finitely generated field extension. Let R be a k-algebra which is a local complete intersection. We can find a global complete intersection $A = k[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ over k such that K is isomorphic to the fraction field of A, see Algebra, Lemma algebra-lemma-colimit-syntomic. Then $R \to R \otimes_k A$ is a relative global complete intersection. It follows that $R \otimes_k A$ is a local complete intersection by Divided Power Algebra, Lemma dpa-lemma-avramov.

Proof of (B). This is clear because a ring is a local complete intersection if and only if all of its local rings are complete intersections.

Part (C). Let $A \to B \to C$ be flat maps of Noetherian rings. Assume the fibres of $A \to B$ are local complete intersections and $B \to C$ is regular. We have to show the fibres of $A \to C$ are local complete intersections. Clearly, we may assume $A = k$ is a field. Then we may assume that $B \to C$ is a regular local homomorphism of Noetherian local rings. Then B is a complete intersection and $C/\mathfrak{m}_B C$ is regular, in particular a complete intersection (by definition). Then C is a complete intersection by Divided Power Algebra, Lemma dpa-lemma-avramov.

Part (D). This follows by the same arguments as in (C) from the other implication in Divided Power Algebra, Lemma dpa-lemma-avramov. Part (E) is immediate as the condition does not refer to the ground field. □